

Efficient Infrared Solar Cells Employing Quantum Dot Solids with Strong Inter-Dot Coupling and Efficient Passivation

Sisi Liu, Chongjian Zhang, Shuangyuan Li, Yong Xia, Kang Wang, Kao Xiong, Haodong Tang, Linyuan Lian, Xinxing Liu, Ming-Yu Li, Manlin Tan, Liang Gao, Guangda Niu, Huan Liu, Haisheng Song, Daoli Zhang, Jianbo Gao, Xinzheng Lan, Kai Wang, Xiao Wei Sun, Ye Yang, Jiang Tang, and Jianbing Zhang*

Lead chalcogenide quantum dot (QD) infrared (IR) solar cells are promising devices for breaking through the theoretical efficiency limit of single-junction solar cells by harvesting the low-energy IR photons that cannot be utilized by common devices. However, the device performance of QD IR photovoltaic is limited by the restrictive relation between open-circuit voltages (V_{OC}) and short circuit current densities (J_{SC}), caused by the contradiction between surface passivation and electronic coupling of QD solids. Here, a strategy is developed to decouple this restriction via epitaxially coating a thin PbS shell over the PbSe QDs (PbSe/PbS QDs) combined with in situ halide passivation. The strong electronic coupling from the PbSe core gives rise to significant carrier delocalization, which guarantees effective carrier transport. Benefited from the protection of PbS shell and in situ halide passivation, excellent trap-state control of QDs is eventually achieved after the ligand exchange. By a fine control of the PbS shell thickness, outstanding IR J_{SC} of 6.38 mA cm^{-2} and IR V_{OC} of 0.347 V are simultaneously achieved under the 1100 nm-filtered solar illumination, providing a new route to unfreeze the trade-off between V_{OC} and J_{SC} limited by the photoactive layer with a given bandgap.

1. Introduction

As the power conversion efficiency (PCE) of crystalline silicon (cSi) solar cells approaching their theoretical Shockley–Queisser efficiency limit,^[1–2] new strategies are essential for further efficiency improvement. Matching the absorption spectrum of cSi with the solar spectrum (Figure 1a), the short-wavelength infrared (IR) (>1100 nm) photons which occupy approximately 20% of the whole sun's power reaching the Earth's surface are unavailable for cSi solar cells due to its indirect large bandgap of $\approx 1.1 \text{ eV}$.^[1,3] Thus, it is a promising avenue to harvest these high transmissible low energy photons (<1.1 eV) for breaking through the efficiency limitation. Theoretical calculation has predicted that up to 6% extra power points can be appended to the Si-based solar cells via cascading IR solar cell to collect the photons beyond 1100 nm of sunlight.^[4]

Dr. S. Liu, Dr. Y. Xia, K. Wang, K. Xiong, Dr. L. Lian, Prof. H. Liu, Prof. D. Zhang, Prof. X. Lan, Prof. J. Zhang
School of Optical and Electronic Information
Huazhong University of Science and Technology
Wuhan 430074, China
E-mail: jbzhang@hust.edu.cn

C. Zhang, S. Li, Prof. Y. Yang
State Key Laboratory of Physical Chemistry of Solid Surfaces
College of Chemistry and Chemical Engineering
Xiamen University
Xiamen 361005, China

Dr. H. Tang, Prof. K. Wang, Prof. X. W. Sun
Guangdong University Key Laboratory for Advanced Quantum Dot Displays and Lighting
Shenzhen Key Laboratory for Advanced Quantum Dot Displays and Lighting
Department of Electrical and Electronic Engineering
Southern University of Science and Technology
Shenzhen 518055, China

X. Liu, Dr. L. Gao, Prof. G. Niu, Prof. H. Song, Prof. J. Tang, Prof. J. Zhang
Wuhan National Laboratory for Optoelectronics
Huazhong University of Science and Technology
Wuhan 430074, China

Prof. M.-Y. Li
School of Science
Wuhan University of Technology
Wuhan 430070, China

Dr. M. Tan
Research Institute of Tsinghua University in Shenzhen
Shenzhen 518057, China

Prof. J. Gao
Ultrafast Photophysics of Quantum Devices
Department of Physics and Astronomy
Clemson University
Clemson, SC 29634, USA

 The ORCID identification number(s) for the author(s) of this article can be found under <https://doi.org/10.1002/adfm.202006864>.

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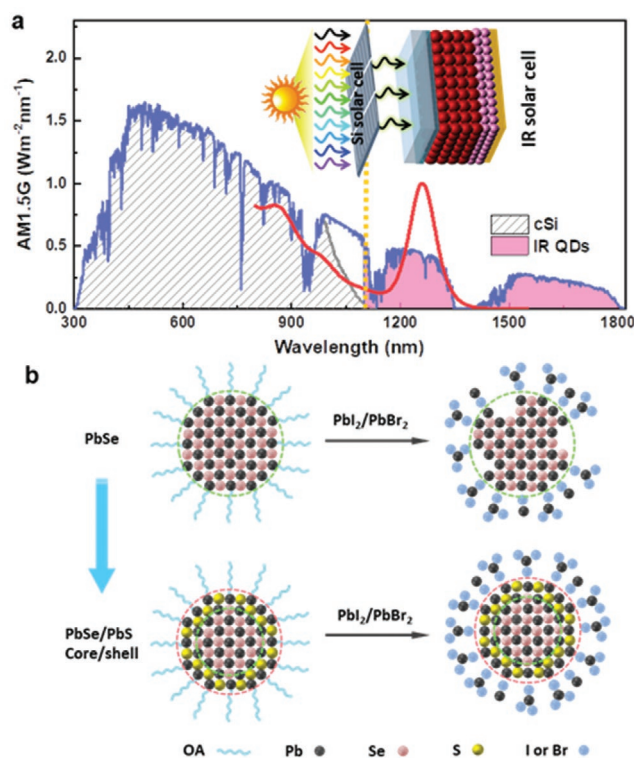


Figure 1. a) The match between AM1.5G spectrum (blue line) and the absorption spectrum of cSi solar cell (dash area). b) The PbS shell protects PbSe QDs during the ligand exchange. The surface of PbSe QDs are not robust enough and might suffer from etching and poor passivation in the ligand exchange, resulting in a relatively high V_{OC} deficit. The introduction of PbS shell protects PbSe QDs, providing excellent trap-state control due to the better air-stability and ligand exchange engineering of PbS QDs.

At present, the IR solar cells mainly employ PbS quantum dots (QDs) as the photon absorber due to their solution processability, strong quantum confinement, and widely size-tuned bandgaps, especially their unique high absorption characteristic in IR spectrum region.^[5–7] The IR efficiency (Si-filtered) of PbS QD based solar cells has reached 1.17%^[8] by improved surface passivation, and was further pushed to 1.34%^[9] via IR light trapping using a nanostructured back reflector. However, the photovoltaic performance is still limited by the intrinsic characteristics of low electronic coupling and carrier transport in PbS QD solids, deteriorating the photocarrier diffusion and extraction and eventually resulting in a low short circuit current density (J_{SC}). Interestingly, the PbSe QDs perfectly compensate these drawbacks in PbS QDs film in the light of the stronger quantum confinement and wave function overlap between adjacent QDs, guaranteeing the excellent carrier transport properties. Indeed, much higher carrier mobilities, up to $24 \text{ cm}^2 \text{ V}^{-1} \text{ s}^{-1}$,^[10] was obtained in PbSe than in PbS QD solids. The better carrier transport should facilitate improved collection of photogenerated carriers in solar cells, which is confirmed by the usually observed higher J_{SC} in PbSe QD solar cells at the bandgap of $\approx 1.3 \text{ eV}$ compared to PbS QD devices with similar thicknesses of the active layers (Table S1, Supporting Information).^[11–15] Thus, it is a potential approach to utilize PbSe QDs to address the low J_{SC} of QD IR solar cells caused by the low

carrier transport of PbS QD solids. In addition, the PbSe QDs have more efficient multiple exciton generation effect than PbS QDs,^[16–18] which further indicates the possibility of driving the PCE of solar cells beyond the Shockley–Queisser thermodynamic limit by introduction of PbSe QDs.^[19] However, there is no report on low bandgap PbSe QDs ($< 1.1 \text{ eV}$) applied in IR solar cells, which can be attributed to the extremely difficult surface passivation engineering caused by the increased non-polar (100) facets and high chemical activity of Se atoms. The inferior surface trap passivation could give rise to high energy disorder and V_{OC} deficit,^[14–15,19] which also is the major reason accounting for the poor PCE of PbSe QD solar cells at the bandgap of $\approx 1.3 \text{ eV}$ regardless of the excellent carrier transport. Therefore, for the QD IR solar cells, the critical challenge for achieving high efficiency is to realize simultaneously excellent carrier transport and efficient trap passivation of the QD solids.

To address this problem, we developed a strategy via combination of epitaxially coating a thin PbS shell over the PbSe core QDs and in situ halide passivation in the present work. Cation exchange synthesis^[3] was adopted to grow the PbS shells with in situ halide passivation that protects the PbSe QDs, giving rise to excellent trap-state control and simultaneously obtaining the superior strong inter-dot coupling and carrier transport. As a result, the PbSe/PbS QD IR solar cells focusing on a bandgap of $\approx 0.95 \text{ eV}$ produce a high V_{OC} which is 1.46 times that of pristine PbSe QD devices due to the improved surface passivation. Furthermore, the PbSe/PbS QD devices demonstrate a much higher IR J_{SC} compared to the previously reported PbS QD devices^[1,3,4,8,9,20,21] due to the strong electronic coupling among adjacent QDs. Consequently, we achieved a high Si-filtered PCE of 1.24% without any light trapping nanostructure. This is the first time that PbSe-based QD solar cells outperform the devices employing PbS QDs, suggesting the great potential of PbSe/PbS QDs for applications in optoelectronic devices.

2. Results and Discussion

PbSe/PbS QDs were controllably synthesized via an advanced cation exchange approach, as shown in Figure S1, Supporting Information. Specifically, CdSe QDs were used to grow the PbSe cores with desired sizes, then small ZnS QDs were injected dropwise for the growth of PbS shells to achieve exciton peaks of $\approx 1250 \text{ nm}$. The epitaxially coated PbS shell with perfect lattice match efficiently protected the PbSe QDs from etching and poor atomic passivation during the ligand exchange, as schematically exhibited in Figure 1b. In addition, the as-synthesized PbSe/PbS QDs were further passivated with the excellent passivant chlorine for PbS QDs, especially for the large-size (low-bandgap) QDs.^[3] Therefore, the PbSe/PbS QDs combine the advantages of PbSe (strong coupling) and PbS (superior surface passivation) QDs, providing the possibility of simultaneously achieving high J_{SC} and V_{OC} originated from PbSe and PbS QDs respectively for IR solar cells.

To understand the surface protection effect of the PbS shell on the PbSe QDs, a series of PbSe/PbS QDs with various shell thicknesses (or PbSe core sizes) were synthesized, as shown in Figure 2. The absorption spectra show that the 1st exciton peaks of PbSe cores gradually red-shifted from 970 to 1225 nm (Figure 2a). After the epitaxial growth of PbS shells, the 1st

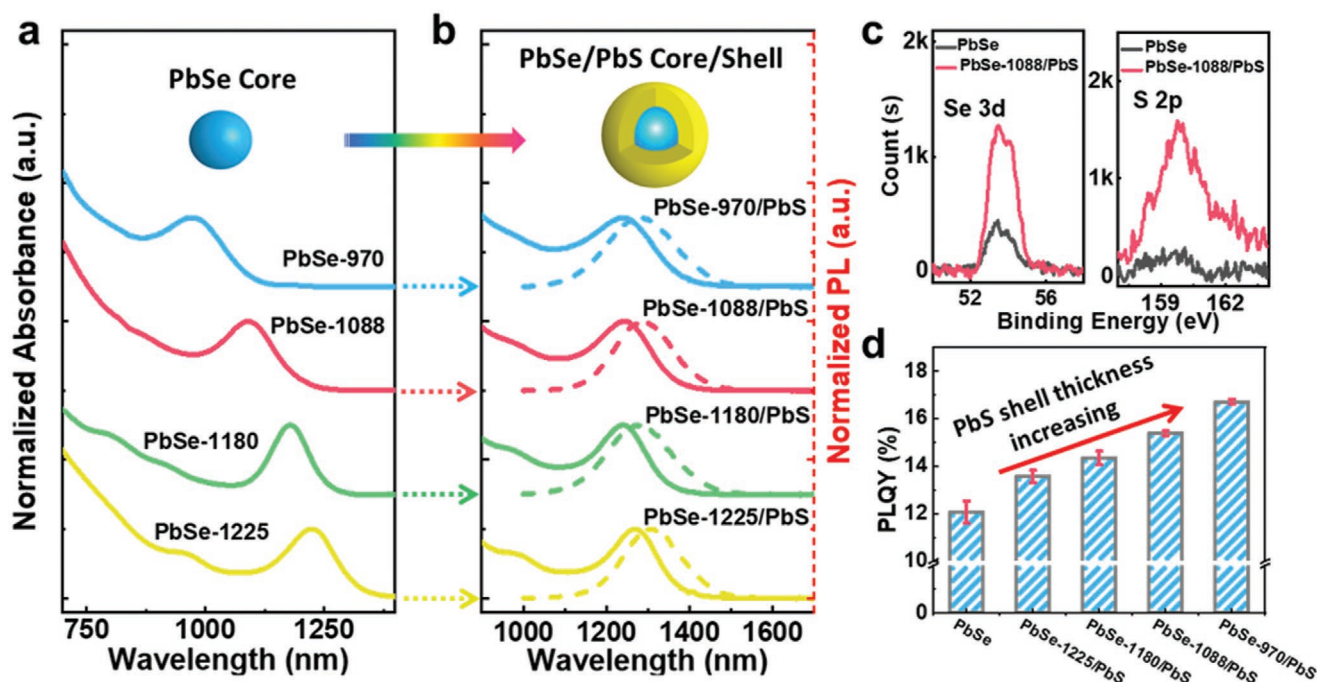


Figure 2. PbSe/PbS core/shell QDs with various PbS shell thickness (PbSe core size). a) Absorption spectra of PbSe core in solutions. b) the absorption and PL spectra of corresponding PbSe/PbS core/shell QDs in solutions. c) XPS spectra of Se 3d and S 2p for PbSe and PbSe-1088/PbS QD samples before ligand exchange. d) PLQYs of PbSe and PbSe/PbS core/shell QDs in solutions.

exciton peaks of the resulting PbSe/PbS QDs are all around 1250 nm, as exhibited with the solid lines in Figure 2b. The shell thickness decreases with the 1st exciton peak of the PbSe core increasing from 970 to 1225 nm. Meanwhile, the similar shapes and peaks of the PL spectra (dashed lines in Figure 2b) for all the PbSe/PbS QDs determined by the radiative recombination of electron-hole pairs from the band-edge transitions indicate the similar size dispersions of these core/shell QDs. As a control, the pristine PbSe QDs with a bandgap of ≈ 0.95 eV were also synthesized, as displayed in Figure S2. X-ray photoemission spectroscopy (XPS) was used to analyze the composition of the PbSe and PbSe/PbS QDs (PbSe-1088/PbS), as exhibited in Figure 2c and Figure S3, Supporting Information. The magnified spectra around the binding energy of the relevant elements reveal the co-existence of both Se and S in core/shell samples while Se was only detected in PbSe QDs, verifying the overcoating of PbSe core QDs with sulfide-based shells. After the epitaxial growth of PbS shell, the photoluminescence quantum yields (PLQYs) of all the PbSe/PbS core/shell QDs in solution are significantly enhanced compared to the pristine PbSe QDs, as shown in Figure 2d. Specifically, the PLQY increases monotonously from 13.5% to 16.7% with the increase of the PbS shell thickness. The enhancement of the PLQY is attributed to carrier confinement as a result of a wide bandgap shell, which has been widely employed in visible emission QDs.^[22] Based on the sizing curves of PbSe^[23] and PbS^[24] QDs, the PbS shell thicknesses of these core/shell QDs are less than 0.72 nm (details can be found in the supporting information). Consequently, the thin shell thickness widens the bandgap of the PbS shell and efficiently protects the PbSe cores, reducing the impact of trap states on the QD surface.

Solution-phase ligand exchange was then employed to achieve electrostatically stabilized PbSe/PbS core/shell QDs before the preparation of photo-active layer. As displayed in Figure S4a, Supporting Information, the oleic acid (OA) capped PbSe/PbS core/shell QDs in the nonpolar octane solvent were replaced by the lead halide ligands, and the QDs were transferred to the polar DMF phase. After the ligand exchange, lead halides were coated on the surface of PbSe/PbS QDs, as demonstrated by the XPS analysis in Figure S4b,c, Supporting Information. The well-resolved excitonic absorption peaks (Figure S5, Supporting Information) and similarly shaped emission peaks (Figure 3a) around 1290 nm for all the halide capped QDs suggest no substantial aggregation occurred during the ligand exchange. Thanks to the coating of PbS thin shells, all the exchanged PbSe/PbS QDs show enhanced PLQYs than the pristine PbSe QDs (Figure 3b), indicating the thin PbS layer indeed improves the trap passivation for PbSe QDs. Additionally, the protective effect of the PbS shell increases with the increase of shell thickness, demonstrated by the monotonous rise of PLQYs of the exchanged QDs in solutions as shown in Figure 3b.

The PL spectra of all QD solids with similar film thicknesses were studied in Figure 3c. The PL intensity rises as the increase of PbS shell thickness. Two factors might contribute to the shell-thickness-dependent PL enhancement. First, a thicker PbS shell provides a better protection on the PbSe QDs, giving rise to reduced surface trap states. Second, a thicker PbS shell leads to a stronger carrier localization, resulting in a higher probability of radiative recombination. Time-resolved PL was used to further investigate the carrier dynamics in the QD solids, as shown in Figure 3d. Following previous reports,^[4,25–26] the PL decays were fitted using a biexponential function and the

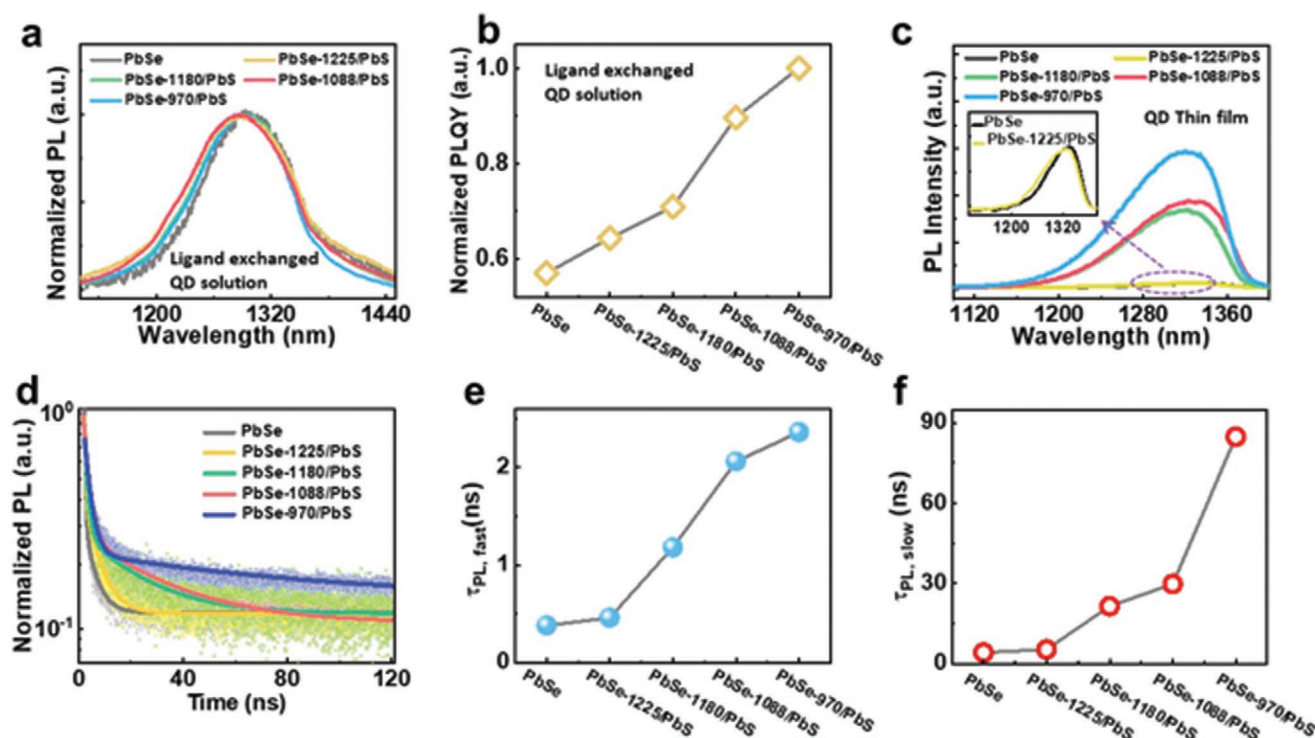


Figure 3. The impact of PbS shell on the surface passivation during the ligand exchange. a) PL spectra and b) PLQYs of PbSe and PbSe/PbS QDs in solutions after ligand exchange. c) PL spectra of PbSe and PbSe/PbS QD solids (the sharp decline in the long-wavelength side is attributed to the detection limit of the spectrometer). d) PL decay of PbSe and PbSe/PbS QD solids. e) Fast and f) slow components of the PL decay of PbSe and PbSe/PbS QD solids. The fast component reflects the inter-dot coupling and the slow component is associated with charge trapping.

dynamics were assigned to two components for electronically coupled QD solids, as presented in Figure 3e,f and Table S2, Supporting Information. The fast decay component is ascribed to exciton dissociation, denoted as $\tau_{PL, fast}$, reflecting the inter-dot coupling. The slow component is attributed to radiative recombination that is limited by charge trapping, denoted as $\tau_{PL, slow}$. The $\tau_{PL, fast}$ increases with the PbS shell thickness (Figure 3e), suggesting the limited inter-dot coupling caused by PbS coating layer. Specifically, the thicker the PbS shell, the stronger the limit. On the other hand, the PbS shell protects the PbSe QDs, resulting in fewer trap states and longer PL lifetimes. Meanwhile, the $\tau_{PL, slow}$ also increases with the PbS shell thickness (Figure 3f), that is, the $\tau_{PL, slow}$ increased from ≈ 4.06 ns for the pristine PbSe film to ≈ 84.9 ns for the PbSe-970/PbS film. The longer $\tau_{PL, slow}$ suggests the reduction of surface traps, verifying the fewer the trap states for the thicker the PbS shell thickness. This is consistent with the shell thickness-dependent PLQY of exchanged QDs in solutions, resulting from the excellent surface protection effects of PbS capping layer.

In addition to the surface trap states passivation, the carrier transport of QDs solids is another critical factor to determine the performance of photoelectronic devices. To evaluate the impact of PbS shell thickness on the inter-dot electronic coupling in QD solids, the photocarrier localization dynamics were examined by time-resolved terahertz (THz) spectroscopy. Stronger inter-dot coupling in QD films can give rise to greater carrier delocalization, caused by the larger wavefunction overlap among the dots, and leading to a larger THz kinetics

amplitude.^[27] The comparison of the THz kinetics in Figure 4a suggests that the inter-dot coupling decreases as the thickness of PbS shell increases. Compared with PbS QDs, a greater portion of the exciton wavefunction in PbSe QDs extends beyond the nanocrystal surfaces due to the stronger confinement,^[28] and thus the PbS QD solid shows smaller inter-dot coupling than the PbSe QD solid. In addition, the decay of the THz kinetics reflects the drop of the photoconductivity, which depends on both carrier mobility and carrier concentration. The normalized THz kinetics (Figure 4b) shows that the decay of the THz kinetics is enhanced as the PbS shell thickness increases. Because these measurements were carried out under the identical pump condition, the QD solids have the same photocarrier concentrations. Thus, the THz decay is attributed to the different carrier localization dynamics, that is, the photocarriers enter localized states faster for a thicker PbS shell. As the delocalized carriers possess increased mobility compared to the localized ones, the enhancement of localization rate for thicker PbS shells may inhibit the inter-dot charge transport. Therefore, the PbS shell reduces inter-dot coupling compared to the pristine PbSe QDs and the PbS QD solid has the weakest coupling.

Based on the above results, on one hand, the PbS shell improved trap-state control of PbSe QD solids, providing the possibility of enhancing the V_{OC} ; on the other hand, the strong inter-dot coupling of PbSe QDs was slightly inhibited by the PbS shell, which might impact the J_{SC} . Therefore, it is possible to achieve a high V_{OC} and J_{SC} simultaneously, and thus produce

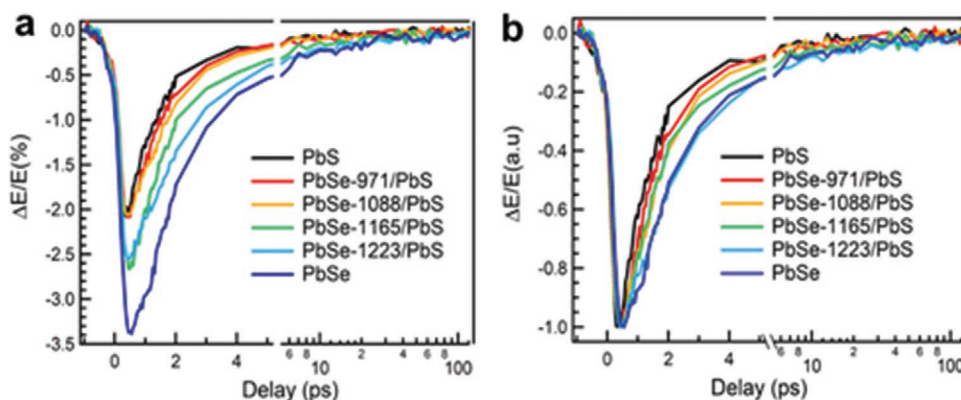


Figure 4. The impact of PbS shell thickness on the electronic coupling in QD solids. The comparison of the a) THz kinetics and b) normalized ones for the indicated samples. The vertical axes represent the relative THz electric field change, and the horizontal axes indicate the delay between the near-IR (800 nm) pump and THz probe, which is split into two parts, -1 – 5 ps and 5 – 120 ps, in linear and logarithmic scale, respectively. The amplitude of THz kinetics represents the ratio of the photoinduced change of the THz electric field to the reference THz electric field.

highest PCE by optimizing the PbS shell thickness. Employing the above PbSe/PbS core/shell QDs, we fabricated photovoltaic devices to harvest low-energy IR photons using an architecture of indium-doped tin oxide (ITO)/zinc oxide (ZnO)/lead halide-QDs/1,2-ethanedithiol (EDT)-PbS/Au, as shown in **Figure 5a**. According to our previous report,^[3] a ZnO film of ≈ 50 nm prepared by sol-gel method was applied as the electron transport layer and a p-type EDT treated PbS QDs (≈ 1.3 eV) film of ≈ 46 nm was utilized as the electron blocking layer, exhibited by the cross-sectional SEM image of the QD solar cell in **Figure 5b**. The photoactive layer has a thickness of ≈ 260 nm. For the IR PCE characterization, a long-pass 1100 nm filter was placed between the solar simulator and the QD solar cells to simulate the four-terminal tandem configuration. **Figure 5c** displays the current density-voltage (J - V) curves of devices based on the pristine PbSe and PbSe/PbS QDs and the parameters are summarized in **Table 1**. The performance of these devices under the

AM1.5 solar illumination is shown in **Figure S6**, Supporting Information. As expected, the introduction of the PbS shell gives rise to a higher V_{OC} due to the better surface passivation and the reduced trap state density than the pristine PbSe QDs, as shown in **Figure 5d** and **Figure S7a**, Supporting Information. For the PbSe/PbS QD devices, the highest IR V_{OC} of 0.347 V was obtained based on PbSe-1088/PbS QDs, which is 46.4% higher than that of the pristine PbSe QD sample (0.237 V) and comparable to many reported IR V_{OC} of PbS QDs based devices.^[29–31] Simultaneously, the fill factor (FF) also increases with the introduction of PbS shell, as exhibited in **Figures S7b** and **S8**, Supporting Information, which verifies the reduction of the trap density in the photon harvesting layer.^[15] The dependency of J_{SC} on the PbS shell thickness under 1100 nm-filtered and AM 1.5 solar illumination is plotted in **Figure 5d** and **Figure S7c**, Supporting Information, respectively. With the increase of the PbS shell thickness, the J_{SC} for the PbSe/PbS QDs maintains

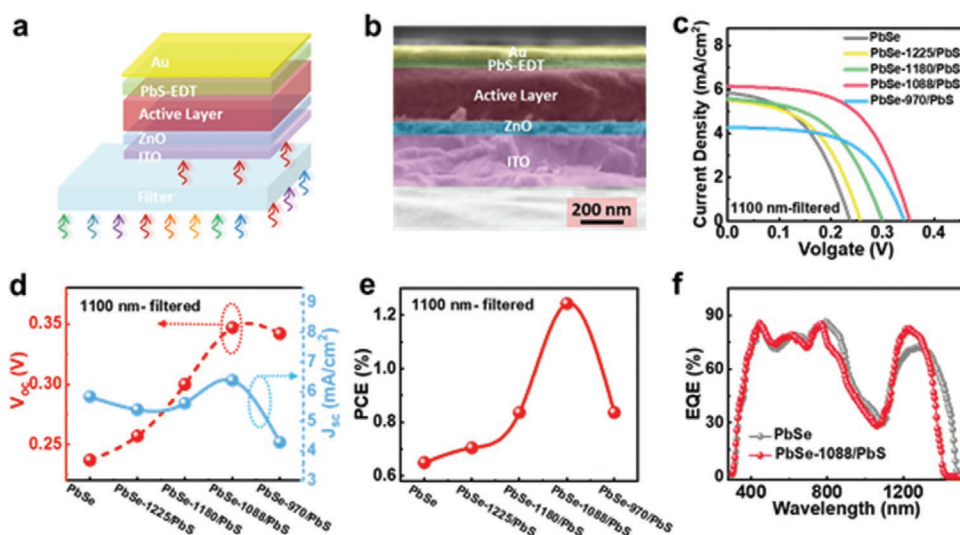


Figure 5. The IR Photovoltaic device architecture and performance under the 1100 nm-filtered solar illumination. a) The illustration of the IR solar cell structure. b) Cross-sectional SEM image of an IR PbSe/PbS QD device. The 1100 nm-filtered c) J - V curves, d) V_{OC} and J_{SC} , and e) PCE of the pristine PbSe and PbSe/PbS core/shell IR QD solar cells. f) EQE curves of the pristine PbSe and PbSe-1088/PbS QD devices.

Table 1. QD solar cell performances measured with or without the 1100 nm filter.

Device	Solar illumination	Voc [V]	J _{sc} [mA cm ⁻²]	FF [%]	PCE [%]
PbSe	AM1.5	0.304	35.4	33.7	3.64
	1100 nm Filter	0.237	5.83	46.8	0.65
PbSe-1225/PbS	AM1.5	0.318	33.3	37.3	3.95
	1100 nm Filter	0.257	5.39	49.5	0.70
PbSe-1180/PbS	AM1.5	0.356	32.3	37.9	4.36
	1100 nm Filter	0.300	5.61	50	0.84
PbSe-1088/PbS	AM1.5	0.405	35.1	43.6	6.2
	1100 nm Filter	0.347	6.38	55.8	1.24
PbSe-970/PbS	AM1.5	0.416	33.3	45.2	6.26
	1100 nm Filter	0.342	4.29	57.2	0.84

a relatively high value that is around that of the pristine PbSe QDs based device. Meanwhile, the carrier mobilities estimated using the space-charge-limited current (SCLC) model for PbSe and PbSe-1088/PbS QD solids are 0.0218 and 0.0208 cm² V⁻¹ s⁻¹ respectively, as shown in Figure S9, Supporting Information, indicating that the introduction of a suitable thin PbS shell has little effect on the carrier extraction. However, further increase of PbS shell thickness resulted in the deterioration of J_{sc} for PbSe-970/PbS QD device due to the inhibition of inter-dot coupling by the thicker PbS shell. Consequently, the largest IR J_{sc} of ≈6.38 mA cm⁻² was achieved for PbSe-1088/PbS QD device. Correspondingly, the PCE of these devices under 1100 nm-filtered solar illumination increases first and then decreases with the increase of shell thickness, as shown in Figure 5e. A superior IR PCE of 1.24% was achieved for PbSe-1088/PbS device, which is 1.91 times higher than that of the pristine PbSe QD solar cell (0.65%). The PbSe/PbS core/shell QDs were used in solar cells previously; however, they failed to exploit the advantages of PbSe-based QDs, that is, the efficiencies were substantially lower than that of PbS QDs based devices.^[32–33] Herein, we show that the performance of PbSe/PbS QD solar cells surpasses that of PbS QD devices, as shown in Figure S10, Supporting Information, and the 1.24% IR PCE represents a high-level efficiency for QD IR solar cells without any light trapping nanostructure.^[1,3,9] The superior performance of the PbSe/PbS QD devices can be attributed to the optimizing of the PbS shell thickness and the in situ halide passivation resulting from the cation exchange synthesis.^[3] The external quantum efficiency (EQE) spectra of the PbSe-1088/PbS and pristine PbSe QD are shown in Figure 5f. A significantly enhanced peak IR EQE (beyond 1100 nm) of the PbSe-1088/PbS QD based solar cells was obtained (82.7%), in contrast to the pristine PbSe QD devices with a peak IR EQE of 72.3%. Moreover, the PbSe/PbS core/shell QD based devices also demonstrate an outstanding stability in air, as shown in Figure S11, Supporting Information.

The significant reduction of the surface trap states in PbSe/PbS core/shell QDs compared to the pristine PbSe QDs (verified by the PL spectra and the PL decay) effectively suppresses the recombination of photogenerated carriers, reducing the V_{oc} deficit of photovoltaic devices. The significantly increased V_{oc} for PbSe/PbS core/shell QDs suggests the importance of PbS shells on the surface passivation of PbSe cores. Meanwhile,

a controllable thin PbS shell coating over the surface of PbSe core could still retain the superior inter-dot coupling and carrier transport of the PbSe QD films (as confirmed by the results of THz spectroscopy and the SCLC), which is beneficial for carrier extraction and a high J_{sc}. Thus, QD solar cells with high V_{oc} and J_{sc} are realized via epitaxially coating a thin PbS shell over PbSe QDs because of both the superior passivation of surface trap states and excellent carrier transport.

3. Conclusions

In summary, we reported an effective strategy to fabricate high-performance IR solar cells (>1100 nm) via epitaxially coating a thin PbS shell with in situ halide passivation over PbSe QDs, which simultaneously realizes strong electronic coupling and superior trap passivation. As demonstrated by the steady-state and transient PL characterizations, the protective effect of the PbS shell depends on its thickness, that is, the thicker the PbS shell, the fewer the trap states on QD surface. Meanwhile, the time-resolved THz spectroscopy demonstrates the dependency of the inter-dot coupling on the thickness of PbS shell. Consequently, at an optimal PbS shell thickness (PbSe-1088/PbS QD), maximum V_{oc} and J_{sc} are achieved simultaneously for the devices based on PbSe/PbS core/shell QDs. The PbSe-1088/PbS QD solar cell exhibits a high IR PCE of 1.24% under 1100 nm-filtered solar illumination, which is a record for IR QD (≈0.95 eV) solar cells without any light trapping nanostructure. Compared to the pristine PbSe QD devices, the prominent improvement of the PbSe/PbS QD based devices is the significantly enhanced V_{oc} due to the effective trap-state control by the PbS shell. In contrast to PbS QD solar cells, the PbSe/PbS QD device shows a higher J_{sc}, thanks to the stronger inter-dot coupling in the PbSe-based QD solids. Therefore, the present strategy allows decoupled defect passivation and charge transport, demonstrating the potential to promote the Si-filtered PCE towards the additional 6% theoretical limit by integrating the intrinsic advantages of PbSe and PbS QDs.

4. Experimental Section

Chemicals: OA (90%), 1-octadecene (ODE, 90%), dimethylformamide (DMF, 99.8%), EDT (98%), and selenium powder were purchased from Alfa Aesar. Lead chloride (PbCl₂, 99.9%), lead bromide (PbBr₂, ≥98%), lead iodide (PbI₂, 99%), oleylamine (OLA, 70%), trioctylphosphine (TOP, 90%) were purchased from Sigma-Aldrich. Cadmium oxide (CdO, 99.99%), thioacetamide (≥99%), zinc stearate (10–12%), diphenylphosphine (DPP, 95%), *n*-octylamine (≥99.5%) and butylamine (BTA, 99.5%) were purchased from Aladdin. Ethanol (≥99.7%), hexane (≥97%), octane (≥98.0%), acetonitrile (≥99.8%), and acetone (≥99.5%) were purchased from Sinopharm Chemical Reagent.

CdSe QDs Synthesis: CdSe QDs were synthesized according to the previously reported method.^[34] In detail, 1.536 g CdO was dissolved in a mixed solution including 96 mL ODE and 11 mL OA, and the mixture was degassed at 80° C for 20 min with vigorous stirring. Then, the mixture was further heated to 250° C under N₂ environment and maintained at this temperature for 20 min to obtain a transparent solution. The solution was cooled to 30° C, and 110 μL of 1 M DPPSe dispersed in 2 mL ODE and 0.57 mL of 1 M TOPSe were successively injected into the mixed transparent precursor. Subsequently, the reaction solution was reheated to 155° C and rapidly cooled to room temperature

by being immediately transferred into water bath. The resulting CdSe QDs were cleaned with ethanol and precipitated via centrifugation, and this process was repeated twice. Finally, the CdSe QD precipitate was dried and dispersed in ODE.

PbSe QDs (≈ 1250 nm) Synthesis: 4.587 g PbCl_2 and 55 mL OLA were added in a three-neck round bottom flask and degassed with stirring at room temperature for 15 min, then heated to 140°C under N_2 and maintained for 30 min. Then the solution was cooled to 80°C , and 5 mL CdSe ODE solution was swiftly injected into the flask. After 10 min, 20 mL CdSe ODE solution was mildly dropped into the flask within 100 min. Next, 90 mL hexane and 40 mL OA were added when the solution was cooled to 40°C in water bath. Finally, the obtained PbSe QDs were washed 2–3 times using hexane/acetone solution and dried in vacuum.

ZnS QDs Synthesis: The preparation of ZnS QDs followed the previously reported method.^[3] 0.45 g thioacetamide, 8.82 g zinc stearate, 100 mL ODE, and 12 mL OLA were added in a flask, and the mixture was stirred and degassed for 15 min at room temperature. Next, the temperature was increased to 140°C and kept at this temperature for 50 min. Then, the flask was switched to water bath for cooling to 40°C and 4 mL *n*-octylamine was injected into the reaction solution. The reaction was stopped once the mixed solution turned colorless. Hexane and ethanol were used to wash ZnS QDs twice, and the products after centrifugation were dispersed in ODE.

PbSe/PbS core/shell (≈ 1250 nm) QDs Synthesis: 4.587 g PbCl_2 and 55 mL OLA were degassed with stirring under vacuum for 15 min in a three-neck round-bottom flask, then heated to 140°C under N_2 and left reacting at this temperature for 30 min to dissolve PbCl_2 . Then, the reaction solution was naturally cooled to 80°C , and 5 mL CdSe ODE solution was promptly injected for the formation of PbSe QDs via cation exchange. After the elapsed reaction time reached 10 min, CdSe ODE solution was dropped into the flask within 35 min to obtain PbSe core QDs with 1st exciton peak at 970 nm. Subsequently, addition of CdSe QDs stopped, and ZnS QDs were dropped into the reaction solution for the formation of small PbS dots via cation exchange. The small PbS dots rapidly dissolved and produced a mass of monomers, realizing the PbS shell growth on the surface of PbSe cores due to their perfect lattice matching. After the 1st exciton peak PbSe/PbS QDs reaching at ≈ 1250 nm, the reaction was stopped immediately. Similarly, for other PbSe/PbS QDs with different dimensions of PbSe core (or thickness of PbS shell), CdSe ODE solution was dropped into the flask within 55, 75, and 95 min to obtain PbSe core QDs with 1st exciton peaks at 1088, 1180, and 1225 nm, respectively. Then ZnS QDs were dropped into the reaction solution to obtain PbSe/PbS core/shell QDs with 1st exciton peaks at ≈ 1250 nm. Finally, the resulting PbSe/PbS QDs were washed 2–3 times using hexane/acetone solution and dispersed in octane.

Solution-Phase Ligand Exchange and Solution Preparation: The process of $\text{PbI}_2/\text{PbBr}_2$ solution-phase ligand exchange was proceeded in air. Precursor solution (PbI_2 0.1 M, PbBr_2 0.022 M) was prepared by completely dissolving the lead halide in superdry DMF. Then, the synthesized oleate-capped QDs octane solution (10 mL, 7 mg mL^{-1}) was added to 10 mL precursor solution and the mixed solution was intensively shaken for 1 min. After the shaking, the QDs in the octane successfully transferred to the polar DMF phase. The DMF phase containing QDs was then purified with octane three times to remove the oleate ligand residues. Next, the solution was centrifuged and the obtained QD precipitate was dried under vacuum for 30 min. Subsequently, the lead halide capped QDs were redispersed in a mixture of BTA and DMF (7:3 in v/v) at a concentration of 320 mg mL^{-1} for the fabrication of photoactive film.

QDs Solar Cell Fabrication: ZnO layer deposited on ITO glass was employed as electron transporting layer (ETL). Specifically, ZnO sol-gel precursor was prepared in the light of previous methods at first.^[3,35] ZnO thin film was deposited by spin-coating the sol-gel precursor on precleaned ITO glass at 4000 rpm for 30 s, and the ZnO film was then thermally annealed at 100°C for 7 min and at 320°C for 13 min. This process was repeated three times to obtain the ETL with thickness of ≈ 50 nm. After the fabrication of ETL, PbX_3^- -capped QDs (320 mg mL^{-1})

in BTA/DMF solution was spin-coated onto ZnO substrate at 2500 rpm for 50 s as the photoactive layer, followed by annealing at 85°C for 5 min in a N_2 glovebox. Then, EDT-exchanged (0.02 vol% EDT in acetonitrile solution) PbS QDs (≈ 1.3 eV) were deposited on photoactive layers by spin-coating in the air, serving as the hole transporting layer (HTL). At last, Au electrodes with effective area of 0.09 cm^2 and thickness of ≈ 60 nm were fabricated on the HTL via thermal evaporation.

Materials Characterization: UV-vis spectrophotometer (Shimadzu UV-3600 plus spectrophotometer) was applied to record the optical absorption of QDs in solutions and solids. The PL spectra of QD solutions were measured using a Zolix OmniFluo fluorescence spectrometer. The PL spectra and decay of QDs solid were collected by an Edinburgh FLS980 fluorescence spectrometer whose detection wavelength limit of the spectrometer was around 1400 nm. The cross-sectional morphology of the device was measured using a GeminiSEM 300 scanning electron microscope. The X-ray photoelectron spectroscopy (XPS) with Al $K\alpha$ excitation (Thermo Fisher, EscaLab 250Xi) was utilized to analyze the elemental compositions of the QDs before and after the ligand exchange.

Time-Resolved Terahertz Spectroscopy: The fundamental pulses were provided by a regeneratively amplified Ti:sapphire laser system (800 nm, 1 kHz repetition rate). The pump pulse was generated by the TOPAS optical parameter amplifier. The THz pulse was generated by optical rectification of the 800 nm pulse (≈ 35 fs pulse duration) in a ZnTe crystal, and then focused into the sample. The transmitted THz pulse was detected by 800 nm pulses in another ZnTe crystal for space electro-optic sampling. The sample was prepared by deposition the QD solid on a quartz substrate and a thin layer of Poly(methyl methacrylate) (PMMA) was used to encapsulate the QD film.

Device Characterization: The dependence of current density and voltage (J - V) was acquired with a Keithley 2400 source under a simulated AM 1.5 (100 mW cm^{-2}) solar spectrum generated from a 450 W xenon lamp (Oriel, Model 9119, Newport). The simulated four-terminal tandem configurations with Si were carried out via placing a 1100 nm (Thorlabs FELH1100) long-pass filter in front of the illumination source during the J - V scanning. EQE spectra were measured by utilizing a Keithley 2400 Source Measure unit and a Newport monochromator. To account for the spectral mismatch between the lamp spectrum and AM 1.5 spectrum, a correction factor was applied to each PCE according to the ratio of the value calculated from integrating the EQE spectrum and the value measured under illumination.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

infrared solar cells, inter-dot coupling, PbSe, PbSe/PbS core/shell, quantum dots

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